MATLAB - SECTION FOUR

Digital Signals in MATLAB

Generating and plotting discrete-time sequences with stem

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BEFORE WE START

What we'll build today

PART 1

Basic sequences

The unit impulse δ(n) and the unit step u(n)

PART 2

Built on u(n)

An exponential and a sinusoidal sequence, switched on by the unit step

PART 3

Shifted sequences

δ(n − 2), a sum of shifted impulses, and u(n − 2)

Tools: ranges with a:b, zeros, for loops, if / elseif, element-wise .^ and .*, and stem plots

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Digital Signals in MATLAB

Part 1: Basic sequences

δ(n) and u(n) · Part 1 of 3

PART 1 · DEFINITION

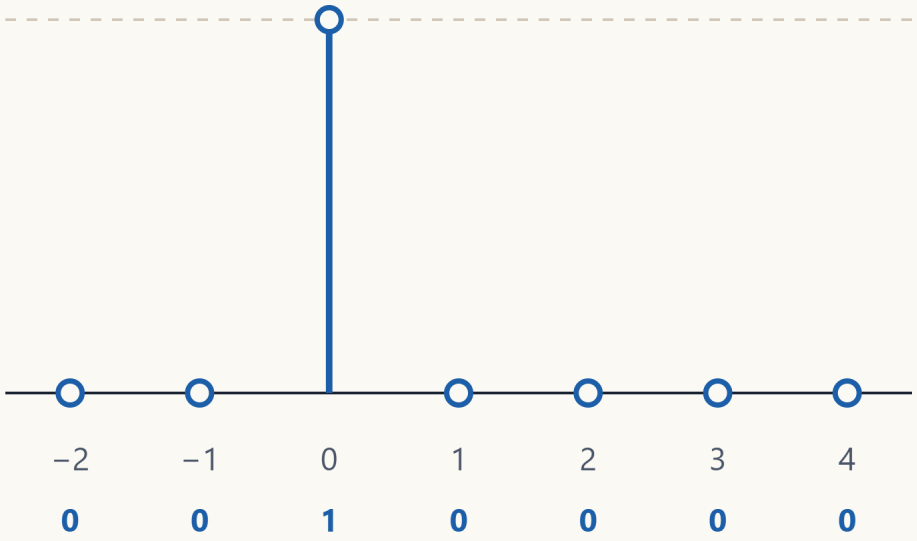
Unit impulse δ(n)

The digital unit-impulse function is a single sample of height **1** at **n = 0**. Every other sample is 0.

DEFINITION

$$\delta(n) = 1 \quad \text{when } n = 0$$

$$\delta(n) = 0 \quad \text{when } n \neq 0$$



Rows: n, then δ(n)

PART 1 · CODE

Drawing δ(n) with a for loop

```
n = -2:4;
s_im = zeros(size(n));
for i = 1:length(n)
    if n(i) == 0
        s_im(i) = 1;
    end
end
stem(n, s_im);
title('Unit Impulse Function');
xlabel('n'); ylabel('\delta(n)');
grid on;
```

How it works

1. **n = -2:4** makes the sample indices −2 to 4.
2. **zeros(size(n))** starts every sample at 0.
3. The loop sets the sample where **n == 0** to 1.
4. **stem** draws each sample as a line ending in a circle.

PART 1 · DEFINITION

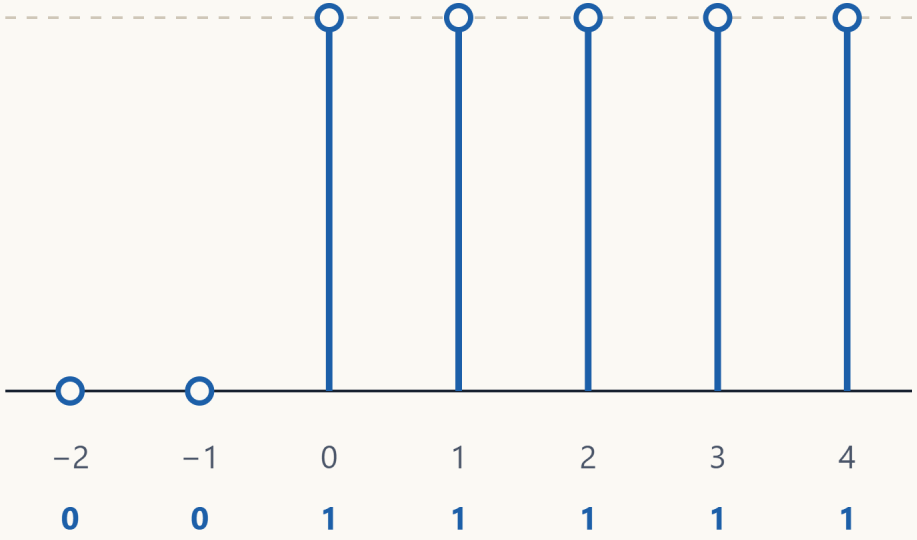
Unit step u(n)

The digital unit-step function is **0** before n = 0 and **1** from n = 0 onward — it “switches on” a signal.

DEFINITION

$u(n) = 1 \quad \text{when } n \geq 0$

$u(n) = 0 \quad \text{when } n < 0$



Rows: n, then u(n)

PART 1 · CODE

Drawing u(n) with a for loop

```
n = -2:4;
u_shifted = zeros(size(n));
for i = 1:length(n)
    if n(i) >= 0
        u_shifted(i) = 1;
    end
end
stem(n, u_shifted);
title('Unit Step Function');
xlabel('n'); ylabel('u(n)');
grid on;
```

What changed

Only the condition. The impulse tests **n(i) == 0** (one sample); the step tests **n(i) >= 0** (every sample from 0 on).

Lab task: run this code, then change the range to n = -5:5 and plot again.

Note: the handout's title line has an extra quote at the end — write title('Unit Step Function');

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Part 2: Sequences built on u(n)

Exponential and sinusoidal examples · Part 2 of 3

PART 2 · EXAMPLE 1

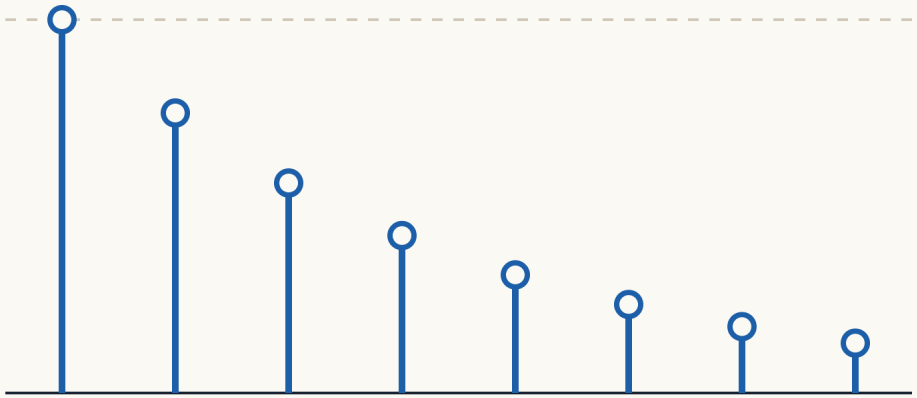
Exponential sequence

THE SIGNAL

$$x(n) = A\,(0.75)^n\,u(n)$$

with A = 10, for the first eight samples n = 0 … 7

Each sample is **0.75 ×** the one before it, so the signal starts at 10 and decays toward 0.



n	0	1	2	3	4	5	6	7
x(n)	**10**	**7.50**	**5.63**	**4.22**	**3.16**	**2.37**	**1.78**	**1.33**

Rows: n, then x(n) (Table 3.2, rounded)

PART 2 · EXAMPLE 1 CODE

Computing $x(n) = 10(0.75)^n u(n)$

```
n = 0:7;
A = 10;
u_n = zeros(size(n));
for i = 1:length(n)
    if n(i) >= 0
        u_n(i) = 1;
    end
end
x_n = A * (0.75).^n .* u_n;
stem(n, x_n); grid on;
title('Exponential Sequence');
xlabel('n'); ylabel('x(n)');
```

Key operators

.^

Raises 0.75 to each value of n, element by element.

.*

Multiplies by u(n) sample by sample, keeping only n ≥ 0.

PART 2 · EXAMPLE 2

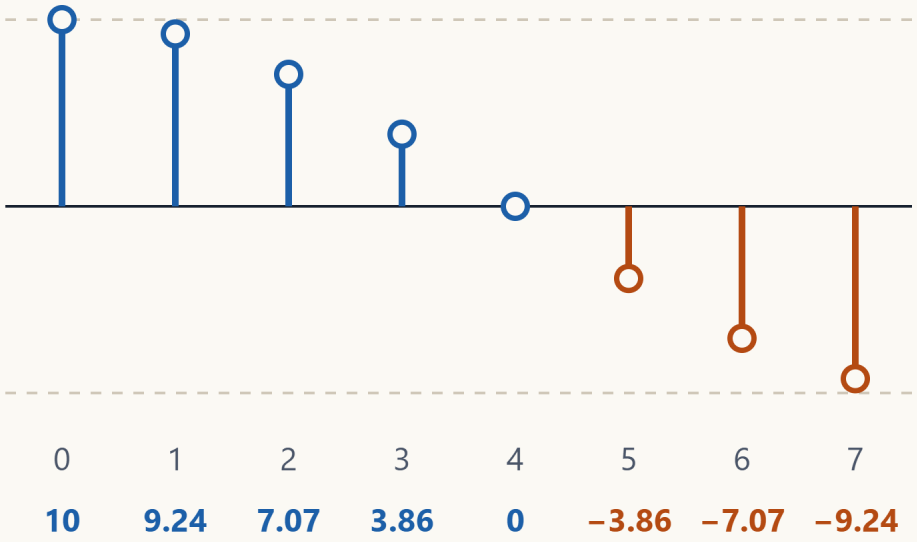
Sinusoidal sequence

THE SIGNAL

$$x(n) = A\cos(0.125\pi n)\,u(n)$$

with A = 10, for the first eight samples n = 0 … 7

The samples follow a cosine: from **10** down through **0 at n = 4** to negative values.



n	0	1	2	3	4	5	6	7
x(n)	10	9.24	7.07	3.86	0	−3.86	−7.07	−9.24

Rows: n, then x(n) (Table 3.1, rounded) · dashed lines at +10 and −10

PART 2 · EXAMPLE 2 CODE

Computing x(n) = 10cos(0.125πn)u(n)

```
n = 0:7;
A = 10;
u_n = zeros(size(n));
for i = 1:length(n)
    if n(i) >= 0
        u_n(i) = 1;
    end
end
x_n = A * cos(0.125*pi*n) .* u_n;
stem(n, x_n); grid on;
title('Sinusoidal Sequence');
xlabel('n'); ylabel('x(n)');
```

Same recipe, new signal

Only the line that computes **x_n** changes from Example 1.

- **pi** is MATLAB's built-in π.
- **cos** works on the whole vector n at once.
- **.* u_n** still switches the signal on at n = 0.

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Digital Signals in MATLAB

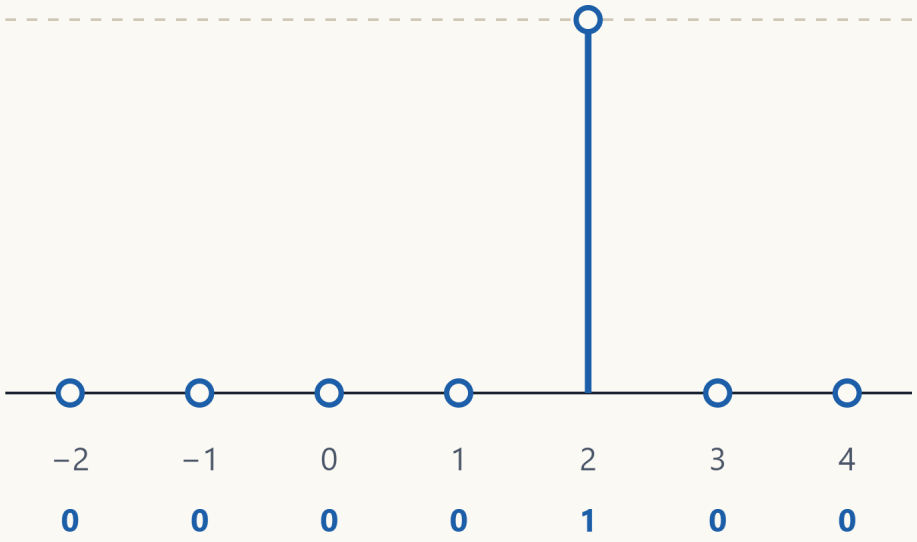
Part 3: Shifted sequences

δ(n − 2), sums of impulses, u(n − 2) · Part 3 of 3

PART 3 · SHIFTED IMPULSE

δ(n − 2): shift to the right

```
n = -2:4;
s_im = zeros(size(n));
for i = 1:length(n)
    if n(i) == 2
        s_im(i) = 1;
    end
end
stem(n, s_im);
title('The shifted unit-impulse');
xlabel('n'); ylabel('\delta(n-2)');
```



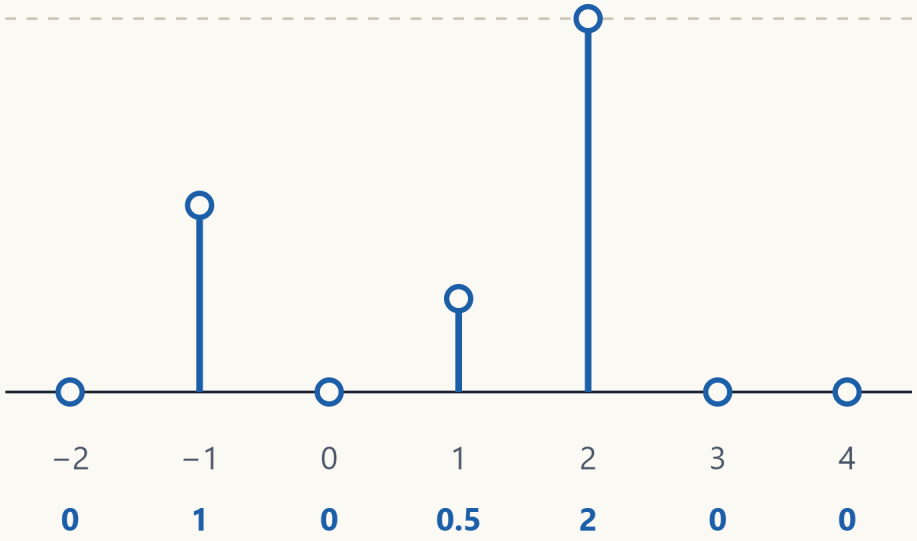
The spike moves from n = 0 to **n = 2**: test **n(i) == 2**.

PART 3 · SUM OF IMPULSES

Combining shifted impulses

```
n = -2:4;
s_im = zeros(size(n));
for i = 1:length(n)
    if n(i) == -1
        s_im(i) = 1;
    elseif n(i) == 1
        s_im(i) = 0.5;
    elseif n(i) == 2
        s_im(i) = 2;
    end
end
stem(n, s_im); grid on;
title('Shifted Unit impuls');
```

x(n) = δ(n+1) + 0.5δ(n−1) + 2δ(n−2)

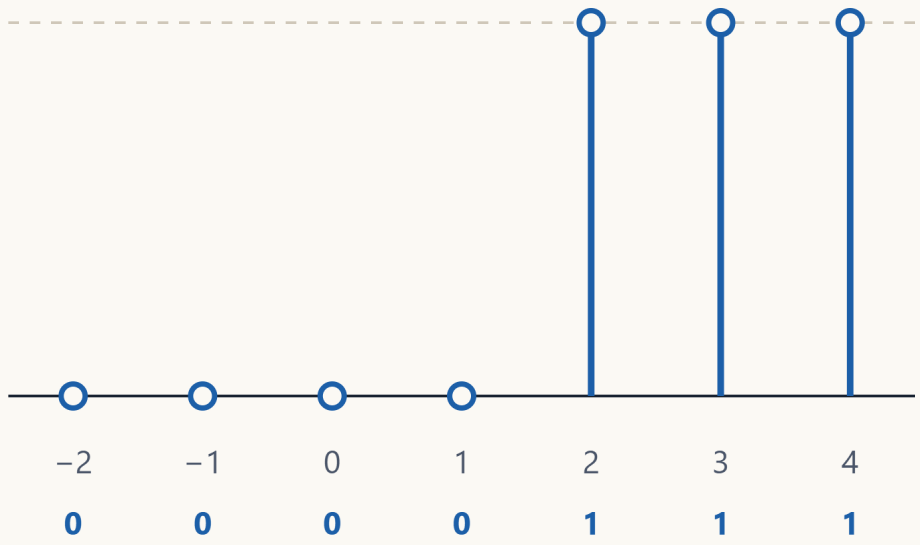


Each term places one weighted spike · dashed line = 2

PART 3 · SHIFTED STEP

u(n − 2): the step starts later

```
n = -2:4;
u_n = zeros(size(n));
for i = 1:length(n)
    if n(i) >= 2
        u_n(i) = 1;
    end
end
stem(n, u_n);
title('Unit step Function');
xlabel('n'); ylabel('u(n-2)');
grid on;
```



The step now switches on at **n = 2**: test **n(i) >= 2**.

THE SHIFT RULE

Minus moves right, plus moves left

x(n − k) · RIGHT BY k

δ(n − 2)

Spike lands at n = 2. In code: **n(i) == 2**

x(n + k) · LEFT BY k

δ(n + 1)

Spike lands at n = −1. In code: **n(i) == -1**

Quick check: set the inside of the brackets to zero and solve for n.

SUMMARY

MATLAB tools from this section

Code	What it does
n = -2:4	Sample indices from −2 to 4
zeros(size(n))	A vector of zeros the same size as n
for i = 1:length(n)	Visit every sample, one at a time
if / elseif / end	Pick which samples get a value
.^ and .*	Element-by-element power and multiply
stem(n, x)	Plot a digital signal as stems

FURTHER PRACTICE

Try these on your own

TASK 1

$u(n + 1)$

Plot it for n = -3:5. Where does the step start?

TASK 2

$5(0.5)^n u(n)$

Plot it for n = 0:9. Does it decay faster than Example 1?

TASK 3

$2\delta(n) - \delta(n - 3)$

Plot it for n = -2:5 using if / elseif.

Tip: start from the closest example code and change only the range, the condition or the formula.

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Thank You

Any questions?

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